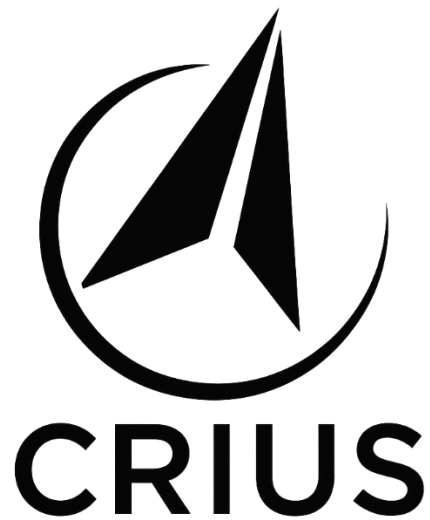




Crius space San-Sat team 2025/26 Preliminary Design Review



Team Name: Crius Space

Country: Poland



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1 INTRODUCTION

CHANGELOG

1.1.1 PDR:

- 1.1
- Secondary mission was slightly changed – We removed pitot's tube from the project, because during our research we found out it would be too big for Can-Sat requirements. However, we are still focused on the movement of our Can-Sat, we want to examine the acceleration influence on the payload in our case mushrooms.
 - We unfortunately had to let go one of the members, because of some personal problems, but we resolved that issue and found a replacement.

Team organization and roles

- 1.2 Our team consist of the 6 student's and 1 teacher; everyone is from the same school PZSP Legionowo (Powiatowy Zespół Szkół Ponadpodstawowych im. Jerzego Siwińskiego). We took initiative to find people capable of coping with various challenges like creating and simulating work for a space company.

Each description was crafted independently, reflecting our unique perspectives and approaches to the same task. This diversity is crucial for technical projects, as it highlights our varied contributions and insights.

1.2.1 Janusz Lejtan (41y.o) (Mentor)

- Multiple fields of expertise such as: programming, graphics, military , psychology, robotics and much more. Graduate of WSPiA University of Lublin in Computer science. 13 years of hands on experience in multiple projects including programming of military drones. Teacher of computer science in Legionowo PZSP school where it is my goal to go beyond standards of public education.

1.2.2 Jakub Wieczorek (19 y.o) (Team Leader)

- Interested in embedded electronics, programing and mechanical engineering, (link to the GitHub profile: <https://github.com/hedfd434>)
- Student of renewable energy in 5th class of the high school.
- Responsible for Electronic/programming/mechanical design of the Can-Sat, GitHub organization,



and documentation.

- Responsible for: managing tasks, Can-Sat hardware and software
- hours committed at the school (weekly): 0h (individual learning program)
- hours committed beside school (weekly): 30h

1.2.3 Roman Lesiuk (15 y.o) (Team Member)

Interested in programming and painting.

Student of information technology (IT technician) in 1st class of high school

Responsible for:

- Creating ground station software and hardware integration.
- Team visual designer.
- Team researcher.
- hours committed in school = 0, in home = 5/6h (weekly)

1.2.4 Grzegorz Rosoliński

- Age: 19 years old,
- Studying IT, year 5th, aspirations to study mechatronics in the future,
- Interested in programming and acquiring knowledge in different fields,
- Responsible for social media, documentation, ground station software, and structural design choices.
- Time spent on the project 4-6h weekly.

1.2.5 Aleksander Dobosz

- ¶ Age: 16 yo
- ¶ Student of the 4th year, IT manager and lecture host at PZSP Legionowo, mikrotik's networking engineer
- ¶ Responsible for
 - coordinating documentation such as readmes
 - helping with editing the PDR
 - Theoretical help with the design of Can-Sat communication module
- ¶ Desired commitment time: None
- ¶ Real time spent doing the project 4-6 hrs. a week.

1.2.6 Tomasz Leszczyński

- Age: 18 years old
- Studying their 5th year as an IT Administrator at a Technical High School
- Responsible for:
 - Assistance with Social Media
 - Can-Sat recovery system (Parachute)
- Time spent weekly on the project: 5-7 hours, mostly spent at home.
- Interests: History, especially the Cold War period. This includes the Space Race and humanity's first rocketry programs.

Mission objectives

1.3 „Space for the mushrooms”

Our mission objective is to investigate the influence of high, rapid acceleration and other factors experienced by the payload during the start of the rocket and descend into the atmosphere similar or identical to the Earth's one, focusing on the reproductive capabilities of the chosen mushroom species.

We would like to use advanced telemetry to monitor our Can-Sat movement and exert forces of the acceleration by (IMU (inertial measurement unit), GPS (Global Positioning System), and barometer readouts for the height changes). To monitor the biological factors, we would like to use temperature and humidity sensors. (In case of real mission, we would use a “cosmic radiation sensor”, in our can-sat we are imagining that we have this sensor)

To expand the existence of humankind we must be able to carry reproductive capable species for the furthest part of space. Food materials and oxygen coming from the plants are the key to humankind expansion.

Our mission goal would be considered a success, when:

- We found our Can-Sat intact.
- We successfully extract and compare our samples with the ones still on land.
- one week after the landing we assess the damage for the mycelium's sprouting capability.

Mission objective in short:

Rocket and space environmental factors influence mushroom reproduction capabilities.



2 CAN-SAT DESCRIPTION

Mission overview

The mission is structured in three stages:

- 2.1 1. Pre-Launch: Select and cultivate *Agaricus subrufescens* mycelium, measure growth rates, and validate its resilience under controlled environmental conditions.
2. Launch: Expose mycelium samples to rocket launch stresses (acceleration, vibrations, microgravity, pressure changes) within the Can-Sat payload.
3. Post-Launch: Analyze the reproductive capacity (spore viability and germination) of flown samples compared to controls, assessing impacts on growth and nutrient potential.

2.1.1 Pre-launch

2.1.1.1 Choosing right mycelium species

Vulnerability of different factors:

The first thing we need to consider is the biological resilience of the mushroom. It must have ability to regenerate after harsh conditions, like vibrations and low temperatures, which will help it throughout the Can-Sat trip at a high altitude.

The sample also has to survive the high acceleration with vibrations and moderate pressure.

Growth consideration:

We would also like it if the mycelium were fast to grow and reproduce, since we do not have much time after the collection of Can-Sat to get the results (Has to grow within a week).

Minimal needs:

The mass and the needs of the sample must be low since in space we would not have that many resources like water, oxygen and other elements required to grow a plant.

2.1.1.2 Comparison of samples (experiments):

We must develop the test and procedures that we will use to grow mushrooms. After the experiment, we will take the sample and compare it with the one we will grow on the ground. Then we will set them up with the same conditions to grow. Later we will check how it grows compared to the one on the ground.

The ideal environment that we will try to achieve is:

- **21-24°C**

- constat humidity. (>75%)
- darkness

2.1.1.3 Initial draft

The first example that caught our attention is **Brown Mushroom (Agaricus bisporus)**. We chose this because of its perfect nutritional content:

- Calories: **3g protein, 4-5g carbohydrates,**
- **vitamin D** and minerals: **selenium, zinc, potassium.**

The mushroom also has great resistance to vibration and is susceptible to microbes. The conditions of growth also align with our requirements. It needs: the temperature of **22 to 25°C, 85-95% humidity**. It also has a short period of partial growth (**<7days**).



Image 1. "brown mushroom, Agaricus subrufescens"

2.1.2 Launch

2.1.2.1 Mission progress over time

For the whole launch we will be:

From launch till releasing the Can-Sat capsule and throughout the descending, including landing of the capsule, we will be taking continuous measurements of **temperature, pressure, humidity, and accelerations** of the payload.

Table 1 "predicted energy usage during start"

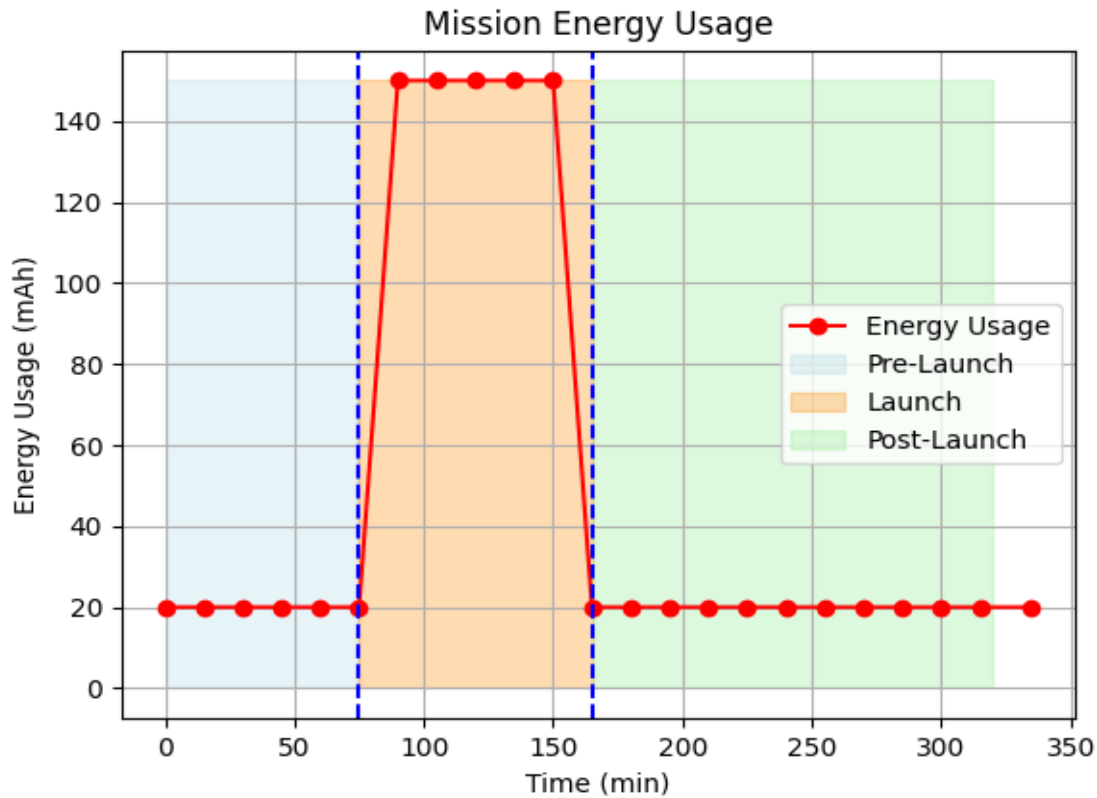


Chart above shows **energy usage**, which we will use with a hibernation mode to preserve energy during stages where we do not need the full computing power. By that we are extending maximum Can-Sat lifetime, which increases our chances of finding Can-Sat and makes it certain that waiting on the launchpad will not be harmful for our mission.

2.1.2.2 Misson stages (from launchpad to finding can-sat in points):

Table 2 "Mission stages"

ID	TEXT	DESCRIPTION
1	sat and waiting on the launch pad	Can-Sat is hibernating (doing everything with the previously prepared procedures)

2	Mission initialization	by sending a message from ground station or through sensors interrupt
3	Launch	our Can-Sat will be ascending inside the rocket
4	Aerial deployment	Our can sat will be automatically deployed at the highest point
5	Descending	parachute will deploy automatically by the air, or it will be triggered by measurement sensors.
6	Landing	shutting down unnecessary systems to preserve energy. (leaving only systems which are required to find can-sat GPS, trans-receiver)
7	Finding can-sat	by tracking the radio signal.

2.1.2.3 Risk management (what can go wrong):

- **harsh weather** – long waiting on the launchpad and gusts of wind that may redirect rocket or the Can-Sat.
- **landing into the trees**
- **bird strikes**
- **mycelium possible issues** – uncontrolled growth (improper storage), sample contamination, mechanical damage before launch.

2.1.3 Post launch

After retrieving the Can-Sat, we will take the mycelium and put it into a controlled environment that we previously set up. Now that we have 2 samples in the same conditions, we will track further growth of both. Based on previous procedures we will compare how differently the one that was in the Can-Sat behaves and we will summarize the experiment.

2.1.4 Required Sensors and devices for the primary mission and secondary mission:

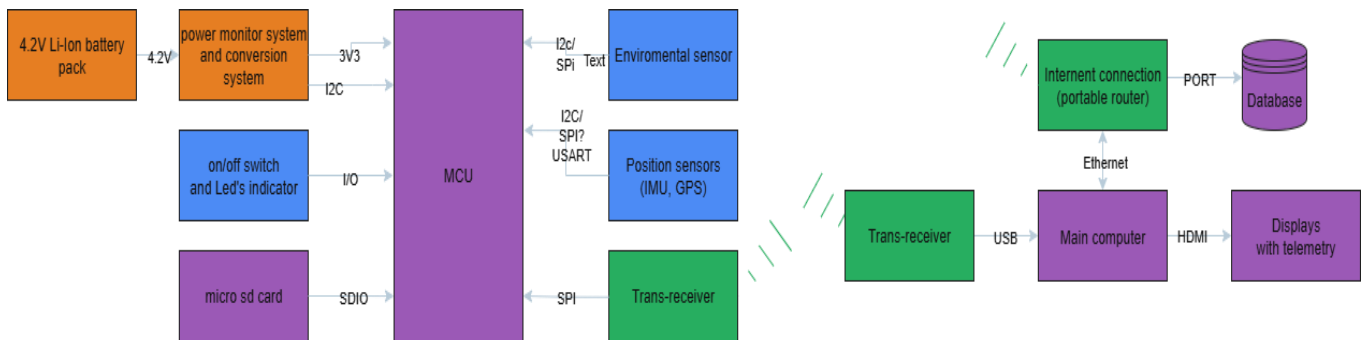
Table 3 “Sensors”

ID	NAME	DESCRIPTION
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1	Environmental sensor and I/O control	which will be monitoring temperature pressure and humidity . as well as working as the input and output (Primary and secondary mission)
2	IMU sensor set	to monitor accelerations and calculate position . (Secondary mission)
3	GPS module	To monitor current position and validate IMU readouts (Secondary mission)
4	Communication module	to provide two-way communication (data exchange and Can-Sat control). (Primary and secondary mission)
5	Flight computer and data processing elements	Data processing , sensors and modules communication. (Primary and secondary mission)
6	non-volatile memory	data storing . (Primary and secondary mission)
7	Power system	Energy distribution and power conversion (Primary and secondary mission)

Diagram 1.



legend:

- sensors -computer and memory -communication modules -power

Mechanical/structural design

2.2.1 Parachute and aerodynamics

2.2

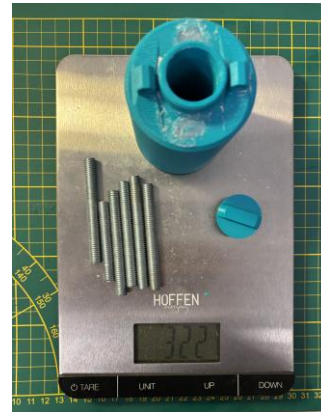
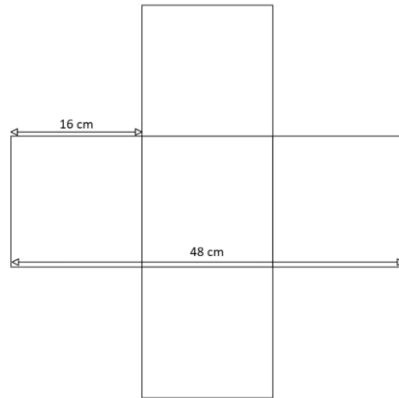


Image 2. "Parachute plan", Image 3. "Can-Sat prototype"

- we have an initial design of our parachute (link below):

<https://1drv.ms/w/c/6581d6bff9332fe2/EW55iB0LRRRIq9bm6rRzF1cBXaM2IOI63rbO8PMoI70x5w?e=w1Sf11>

- in the future we are considering using a **manual parachute** deployment by servo or **solenoid**.

- aerodynamics of Can-Sat – Can-Sat will have a uniform cylinder shape with a smooth surface that will not be causing any **air distortions**.

Link to the GitHub repository with first project of the can-sat body that includes parachute mounting: https://github.com/hedfd434/can_sat_initial_models

2.2.2 Main body material

Our initial idea is to create a body/structure of Can-Sat using 3d printing technology (using **PLA**, because it is widely available, relatively cheap, strong easy to work with), it will enable us to easily project our models in **CAD programs (like Fusion 360)** and print it.

2.2.3 Compartments:

We want to divide the structure of our Can-Sat into a few sections:

- **Parachute section** (parachute and/or deployment system.)
- **Biological payload** (mycelium samples.)
- **Computer and sensors section** (board with main computer and embedded sensors.)
- **Power section** (batteries and power converter.)

Initial mass balance:

Table 4 "Mass balance"

ID	NAME	METHODE AND DETAIL	WEIGHT
1	Net mass of the body	(based on the first body prototype (very dense because of internal structure)) + screw/nuts	110g
2	electronic components	(measured) (2 * 14500 4.2V batteries)	40g
3	power plant	(2 * 14500 4.2V batteries) (measured)	50g
4	parachute + deployment system	(servo/solenoid) (datasheet of propositions)	30g
5	biological payload	(clear assumptions) + (with the accessories)	80g
Total			310g

2.3

Electrical design

2.3.1 General architecture

General Idea says that we will have a **microcontroller (master)**, and a few devices like sensors, drivers and **trans-receivers** (slaves). Sketch of this configuration is described in **Diagram 1 (page 10)**.

In the sections below we want to describe to you our initial drafts and ideas for certain elements. In **table 3 (page 3)** you can see overview of the elements required to build our Can-Sat.

2.3.2 Primary mission devices

For the primary mission we need a board with the computer/microcontroller which will be responsible for data processing and communication with the other modules.



Member of our team have experience in creating custom PCB here is the link to the GitHub repository with one of his latest project, which was a preparation for this competition:

https://github.com/hedfd434/custom_eval_stm32L43KCU6

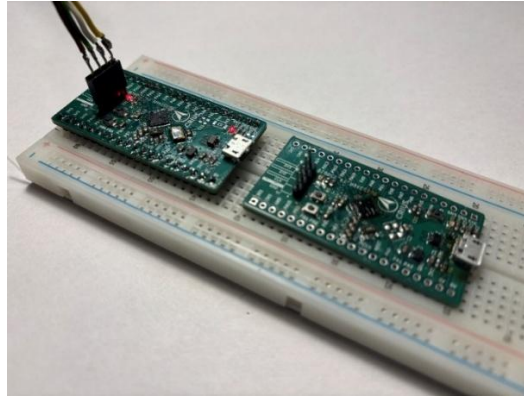


Image 4. "Development board for our project"

We will also need a **temperature**, **pressure** sensor, communication module, and non-volatile memory in form of the **micro-SD card**.

for the temperature pressure and **humidity** sensor we want to use a **BME280** from **Bosch** which is easy to use. (We will also utilize humidity in the secondary mission).

And we have already used in in our previous projects

link to the **GitHub** example of usage: https://github.com/hedfd434/stm32_bmp280_library

2.3.3 Secondary mission devices

To proceed with a secondary mission, we will need an IMU **sensor set**. (We need **9dof** measurements). So far in the previous project we have been using **BMI160** (**6dof - accelerometer + gyroscope**) and **HMC5883L** (**magnetometer**, widely known as a **GY-273**). However, both of those modules are **obsolete**, so that means that they are no longer produced. Despite that many electronics stores have a high stock of them we will consider using other universally available sensors set.

We will also need a **GPS** module; from our previous projects we can say that the **U-Block** products are reliable and easy to work with. So, we will consider one of their modules like **MAX-Q8** or **MAX-Q10**.

For the secondary mission for the sake of opening the parachute manually, we might also need a **solenoid** or a **servo**. For current stage we know that They are proper models to use in our mission (powerful, low energy usage, small size).



2.3.4 Power supply

Our power system initially will contain a power source in form of the **4.2V Li-Ion battery pack** (they are available in many sizes that will fit to our size and power requirements) with the estimated minimal capacity of **1500mAh**. This set will allow us to overcome issues related to waiting on the launchpad and after landing we will be able to find our can-sat for a long time.

On board we will have a power conversion system to provide required voltage for sensors. It will have to be power-efficient and able to deliver required power at the desired parameters (**3.3V** and **500mAh (in peak)**).

For the current stage we can have a rough estimate of power usage based on previous project. we are estimating it on about **120-150mAh** in the “**working mode**” and about **20mAh** in the “**hibernation mode**”. We will not have any components that are required high power usage for a long amount of time (probable solenoid or motor will be working only for a small piece of the mission duration (10-15 seconds)).

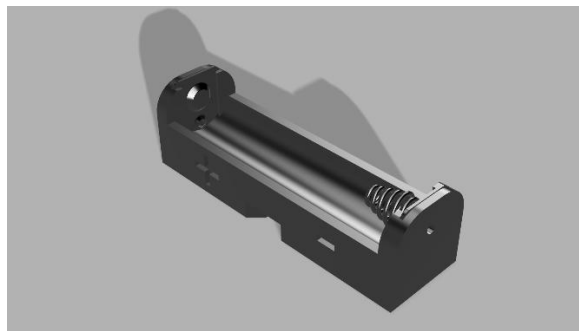


Image 5. “Render of the battery holder”

Link to the **GitHub** repository with the custom battery holder prototype:

<https://github.com/hedfd434/3D-printed-battery-holders-containers->

2.3.5 Communication system

We want to have a **two-way** communication system in our Can-Sat, that will allow us to not only receive data but also send a command to our Can-Sat, this part will show our capabilities in terms of creating devices capable of being controlled from the remote flight center

To align with the requirements of the competition we initially thought of **RA-02** (module based on **SX-1728** that we have already used in many previous projects as low-power reliable medium of communication (module of the official board)).

Software design

2.4.1 Can-Sat software

It will be responsible for data acquisition and further processing as well as for controlling communication modules and potential motor or solenoid.

Language, IDE, technology: **C/C++, stm32CubeIDE, HAL library.**

We will have the trans-receiver on the ground that will be using the same software designs as the Can-Sat + it will be able to communicate with the computer in order to pass acquired data

2.4.2 Ground station software:

Ground station software will be responsible for data **acquisition, processing, saving (locally and remotely using database) and graphing.**

We want to have a live panel with telemetry data:

Language, ID, technology: **Python, C/C++, Unity + environment, Visual Studio code.**

2.5 Recovery system

As our recovery system we want to use a parachute (link to our sub-report:

<https://1drv.ms/w/c/6581d6bff9332fe2/EW55iB0LRRRIq9bm6rRzF1cBXaM2IOI63rbO8PMoI70x5w?e=w1Sf11>) and a **radio signal** with **GPS localization** to locate it after. We will consider using **bright neon colors** and/or **sound/light indicators**.

Ground support Equipment

Table 5 "Ground station equipment"

ID	NAME	DESCRIPTION/FUNCTION
1	Communication module	Used for two-way communication, data collection, and monitoring during the mission.
2	Main computer	Processes incoming data and manages communication software. (+ backup)
3	2 screens (2 backup)	Display real-time telemetry, graphs, and mission parameters. Backup screens are available in case of display malfunction.

4	Spare parts	Additional components and a backup Can-Sat are prepared for emergency replacement or testing before the launch.
5	Software	Custom software is used to receive, decode, and visualize telemetry data from the Can-Sat. It displays parameters like altitude, GPS coordinates, and signal strength in real time.

3 PROJECT PLANNING

Our Team Communication Work System.

- 3.1 Since **Messenger** is accessible on everyone's phone, our team uses it for quick, daily communication. We have a dedicated **Discord server** for meetings and brainstorming. This server includes channels for main discussions, report writing, social media, structural-electrical design, and voice calls. We use **Trello** to manage task assignments, as explained below.

link to our Trello board:

<https://trello.com/invite/b/6702cb28e0af29db2f87c9f1/ATTI03971901b4f231ac8a435d646d27be06923DC768/cansat> (you must **sign in**, it is **free**.)

- 3.2 **Managing Files.**

To ensure you always have access to the latest versions, we keep project files in a **GitHub** repository that contains folders for each project component. We work on the report and parachute sub-report using **OneDrive**, uploading updated versions to **GitHub** after major revisions. Link:

- 3.3 https://github.com/Can-sat-2025-26-Crius-team/main_repo (**public**)

Organizing the Task.

Jakub Wiecezorek, the team leader, assigns tasks on **Trello**, specifying each task for a particular team member. The Trello board has five sections: To Do, In Progress, To Review, Done, and Currently

- 3.4 Done (for tasks completed at the current project stage, since tasks cannot be split into smaller sub-tasks). After finishing, tasks move to "To Review," where the team leader checks them before moving them to "Done," "Currently Done," or back to "To Do" if changes are needed.

Schedule

This is a schedule:

Table 6 "Planned schedule"

ID	PHASE	DESCRIPTION
----	-------	-------------



1	Initial planning (PDR REPORT) (19.10.2025)	In that phase we will develop an initial outline of our mission and project details.
2.1	Technical consultations (3 - 4 November 2025)	We will think of every possible hardware problem and try to solve them; we will also decide on our initial modules for our Can-Sat.
2.2	Main development, first prototypes and mycelium testing (November 2025 – January 2026)	In this stage we are going to develop first prototype of our Can-Sat and ground station all together with first prototypes of the software solutions. We are also going to test our chosen mycelium in variable controlled environment and develop growth testing procedures.
3	Before CDR report (January 2026)	We are going to test simulate a can-sat launch, including Can-Sat and ground station. As a descending stage we can use eg. drones, for ascending stage and durability we can throw our can-sat by hand and measure the acceleration and forces produced by the on board IMU sensors. After that we will put previously prepared "test reports"1 into our CDR
4	CDR Development (18 January 2026)	In that phase We will show you what we have learned from our previous tests and experiments, we will present and explain what we have accomplished so far. And what else is to upgrade or reconsider.
5	After CDR consultations (2 February 2026)	So far we do not know is it going to be, but We are assuming that will be similar in form of to the first meeting with the organizers, where We will discuss our possible technical and other issues.
	Last prep before FDR (March)	Finalization of preparing Can-Sat and ground station modules and software that it will work with. We also expect at this level that we will be certain that our... solution? That will work.

7	FDR development (29 March 2026)	In this step we must summarize all the research we conducted earlier in order to fully show what our mission will look like and what we expect as results from the final launch. The conclusion is also expected to be made at this part.
8	Final stage before the start (April 2026)	In this stage We will know if we are going to launch our Can-Sat during start campaign. In the case of being choose we are going to revise our start plan and list all of the required items (including spare parts for both can-sat and ground station) together with chosen mycelium samples.
9	The start campaign (18-19 April 2026)	During the Start Campaign we will show the organizers our final product of our work and they will all be able to see it in action. During that stage We will perform a "Mycelium for the Space" experiment.
10	OFF-Scene part of the experiment (19-24 April 2016)	During the week after the campaign starts. We will perform previously prepared tests with use of the samples which will be in can-sat during launch.
11	Confirmation or refusal from thesis (19-24 April 2026)	The experiment summarizes and concludes if mushrooms can withstand extreme conditions such as space travel. We are doing this experiment purely for scientific purposes and that part will show us the output of it.
12	Final presentation (24 April 2026)	We will participate in the final presentations when we show outcomes of our experiment and presentation and present solution in refer to the outcomes

3.5

Resource estimation

3.5.1 Budget

Table 7 "Budget rough estimations"

ID	NAME	PRICE
----	------	-------

1	body material (including screws and nuts)	25€
2	electronics	50€
3	sensors	20€
4	Parachute + deployment system	10€
5	Batteries (with minimal total capacity of 1500mAh)	10€
6	mycelium samples	20€
total		135€

3.5.2 External support

Institute of Fluid-Flow Machinery of the Polish Academy of Sciences (PAN):

Using our PDR report we managed to interest a team from Research Center KEZO, Institute of Fluid-Flow Machinery of the Polish Academy of Sciences (PAN), The team expressed their willingness to support us in the implementation of the competition project. Thanks to this cooperation, we can count on invaluable support in conducting, analyzing, and documenting scientific research, as well as access to measuring equipment and research devices.

Link to their website: <https://www.imp.gda.pl>



SARW S.C.:

We have also established cooperation with SARW S.C., experts in the field of automation! Thanks to this, we can obtain professional advice on coding and prefabrication of electronics. In addition, we will use their professional soldering stations — under the supervision of experts! Links to our partners' websites:

Link to their website: <https://www.sarw.pl>



OUTREACH PROGRAMME

3.6.1 Purpose and Communication Strategy

Our goal is to increase the visibility of our project, engage the local community, educational environments, and potential collaborators and sponsors. We aim to build the image of a committed, innovative, and cooperative team.

The communication strategy includes:

- Regular updates about project progress (via social media, blog, vlog).
- Collaboration with local and industry media.
- Contact with creators and experts in the space technology and biology (mycology) field.
- Organizing presentations, demonstrations, and exhibitions.

3.6.2 Digital forms of the content

To show our work to people, we create the following social media accounts:

- Instagram (<https://www.instagram.com/cruiospace/>) where we publish things like:
 - Photos and short videos of the design and construction process of our Can-Sat.
 - Educational posts explaining how different components of the Can-Sat work.
 - Announcements of events, presentations, and school activities related to the project.
- TikTok (<https://www.tiktok.com/@cruiospacecansat>). It is for future content.
We expect to publish educational videos about the competition there.

Social media will allow us to reach a larger audience. This will help us engage more individuals in our field of interest and find potential partners or collaborators.

3.6.3 Promotion Through Local Media

We plan to promote our project through local media and online social media outlets. Our idea is to spread awareness about our Can-Sat project.

3.6.3.1 This outreach will help us:

- Reach people outside our school or in a direct environment,
- Gain support from the local community, sponsors, and educational institutions,
- Inspire other students and young people to get involved in space and technology projects,
- Strengthen our team's visibility and credibility as a part of the national Can-Sat competition.